

Yaswanth Deevi

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SUMMARY

Computer Science Master's Student at BTH with substantial experience in deep learning, computer vision, and processing multi-modal sensor streams. Skilled in transfer learning, model compression (quantization/pruning), and porting machine learning models on heterogeneous architectures/hardware. Familiar with using camera, pose, and raw signal sensor streams and aggregating them in a unified decision-making pipeline.

EDUCATION

M.S. in Computer Science & Engineering

Expected Jul 2027

Blekinge Tekniska Högskola, Sweden

Relevant Coursework: Research Methodologies, Advanced Machine Learning, Deep Learning, Decision Support Systems

B.Tech in Computer Science & Engineering

Dec 2025

Jawaharlal Nehru Technological University Hyderabad

Relevant Coursework: Machine Learning, Data Structures and Algorithms, Computer Networks, Object Oriented Programming, Python Programming.

EXPERIENCE

Computer Vision Engineer (Pose & Interaction)

June 2026 – Aug 2026

SolarFi's FairWAI, Boston, USA (*Full work under NDA*)

GitHub Repository

- Developed perception pipeline that consists of multi-person pose estimation in two drastically different backbones and hardware backends: Vision on iPhone Neural Engine and INT8 quantized MoveNet Lightning on CPU, under identical constraints of accuracy and latency, without starting from zero.
- Built hardware agnostic interaction layer: temporal pose-to-primitives mapping mechanism (geometric heuristics + calibrated MLP), that took keypoint output of both backbones and produced common representation for downstream use.
- Built iterative budgeted textual summary that integrates multi-person pose with interaction state into a unified interface to be consumed by the vision-language model – pretty hard task to integrate multiple sensors inputs into common interface.
- Integrated calibrated confidence information and temporal smoothing such that the signal from either backend would not interfere with the fused representation due to noise.
- Developed a YOLOv8 and solvePnP-based improvement for 3D head-pose estimation to address two interaction primitives not performing well, implemented on top of the current pipeline (verified with macro F1 score 0.853 for 148 labeled videos) through an independent implementation which did not impact the existing system.
- Performed per-primitive precision and recall analysis and skin color/ethnicity bias analysis on a batch of 50 videos for accuracy and fairness verification.
- Worked as part of a 3-person team by providing events and track IDs and having weekly hardware integration meetings (thrice a week).

Research Intern

Jul 2025 – Sep 2025

Defence Research and Development Organisation (DRDO), Hyderabad

GitHub Repository

- Developed and tested a unique approach to solve the P-Center facility location problem (P centers to be selected out of n total nodes minimizing the maximum distance between centers and clients) using a combination of evolutionary Invasive Weed Optimization algorithm with deep neural network fitness function approximation.
- Implemented a novel rank-based probability-based scheme for parent/survivor selection maintaining population diversity and preventing premature convergence in conjunction with self-adaptive Gaussian mutation strategy with logarithmic update law for mutation strength of each solution.
- Demonstrated up to **100x** speedup of the optimization process compared to pure Invasive Weed Optimization (from 371.03s to 3.11s at N=3000 and P=100) at the cost of small decrease in the quality of the solution (e.g., fitness 49 vs 63 at the same problem instance), benchmarked on various problem instances ranging from N=10 to N=3000 nodes and P=3 to P=100 centers on symmetric random distance matrices.
- Developed and tested the surrogate NN training pipeline using the fitness data gathered from the evolutionary runs, which proved that the prediction accuracy increases with the training data volume (correlation coefficient $R^2 \sim 0.12-0.67$ for various instance sizes).
- Created a variant of the Artificial Bee Colony algorithm as another reference point in order to compare the hybrid approach to a different metaheuristic method from the literature.
- Wrote a technical report in which the task was formulated as an integer linear program and a comparative experimental study of the algorithms was conducted.

KEY PROJECTS

Classifier Robustness to Character-Level Noise in Sentiment Analysis *(Research Methodologies Coursework)*

[GitHub Repository](#)

- Conducted a controlled experiment using four classifiers for sentiment classification in the SST-2 dataset: Logistic Regression, Multinomial Naive Bayes, Linear SVM, and Hard Voting Ensemble. The experiments used three character-level noise models with varying levels of intensity, five different random seeds.
- A paired t-test among 15 noise models was performed to find whether the observed difference in robustness is statistically significant. There was no statistically significant difference between the ensemble and Logistic Regression.
- It was found that Naive Bayes suffers from out-of-vocabulary token sensitivity in the case of character deletion noise due to its probabilistic decision function.

Deepfake Detection System *(From Undergraduation Coursework)*

[GitHub Repository](#)

- Used Transfer Learning for the process of fine-tuning the pre-trained EfficientNet-B0 on the problem of binary classification of deepfake videos based on Celeb-DF dataset.
- Designed a video inference pipeline at frame level using MTCNN with dynamic frame skipping.
- Created a complete Flask-based web application and performed analysis of the performance metrics of the solution like F1 score, ROC AUC, confusion matrix, etc. at different compression levels and lighting conditions.

Neural Digital Pre-Distortion (OpenDPD) *(Independent Research Work With Professor)*

[GitHub Repository](#)

- Created two new architectures (MLP with skip connection and BiLSTM with scaled dot-product attention) to implement the same signal processing task utilizing paired IQ sensor data for the purpose of evaluating the feasibility of these architectures in relation to hardware constraints as the firsthand experience of implementing different architectures on the same task.
- Created an indirect learning training pipeline using sliding window sampling of multi-modal signal data, which is applicable to any sequence of sensor data.
- Performed performance evaluation using EVM (%) and NMSE (dB) for both the architectures and explained the reason for BiLSTM's superiority over MLP because of its high effective memory as the comparison criterion.

LSTM Autoencoder for Financial Fraud Detection *(From Deep Learning Coursework)*

[GitHub Repository](#)

- Implemented an unsupervised reconstruction-based anomaly detection pipeline on a PaySim-like dataset containing 6.36M transactions, by taking 10 chronologically sorted transactions as a sliding window and training two types of LSTMs autoencoders (single layer and a bidirectional 2 layers) without using any fraud labels during training.
- Compared both types of autoencoders to the Isolation Forest, where the better performing autoencoder (bidirectional autoencoder) was having the best ranking (ROC-AUC = 0.838), even in case of extremely imbalanced dataset.
- Derived reconstruction error threshold from only training portion of the data and generated diagnostic plots such as time series, scatter plots, and error histograms in order to understand the problem of good ranking but bad precision and recall (at a certain threshold).

Anomaly Detection in Building Heating Systems *(From Advanced Machine Learning Coursework)*

[GitHub Repository](#)

- Created a data anomaly detection workflow using FP-Growth association rules by streaming sliding windows with size 6 and stride 3 that have 50% overlapping to ensure that the detection delay stays under 30 seconds.
- Used weighted scoring with rules filtering along with rules' stability validation through chronological N -fold cross-validation.
- Made the comparison of the obtained results with 3 unsupervised algorithms (Isolation Forest, K-Means and DB-SCAN) by using agreement matrices and ROC graphs.

TECHNICAL SKILLS

Computer Vision & Perception: CNNs, Pose Estimation (Apple Vision, MoveNet), YOLOv8, Face Detection/Alignment (MTCNN), Frame-level Classification, Multi-modal Signal/Image Processing, OpenCV, ONNX Runtime

Deep Learning & Model Optimization: PyTorch, TensorFlow, Keras, RNNs, BiLSTM with attention, Transfer Learning, Model Quantization (INT8/4-bit), Model Pruning, Cross-architecture deployment (Neural Engine & CPU), Core ML

Optimization & Search: Evolutionary Algorithms, Surrogate Modeling, Combinatorial Optimization, Metaheuristics, with a focus on reducing computational cost and hardware-aware trade-offs

Machine Learning & Data: Scikit-learn, XGBoost, LightGBM, PCA, SHAP, Statistical Hypothesis Testing, NumPy, Pandas, SciPy, Matplotlib, Time-Series Analysis

Programming & Tools: Python, C++, SQL, Git, Docker, Linux

CERTIFICATIONS & AWARDS

- Runner-up, PEMA Ottonel Popesco Student Challenge 2025/2026 – reached the finals representing JNTU Hyderabad
- IIT KGP AI4ICPS – Certificate of Merit – Advanced Machine Learning & Industrial Cyber-Physical Systems
- Smart Interviews – Data Structures, Algorithms & System Design